I

Code, Compliance & Health Department

A kitchen does not open because it is built. It opens because it passes. Code compliance is not a final-stage correction. It must be designed into the space from the very first layout.

WATCH AND LISTEN

Section Briefings

Watch the video, listen to the podcast, then read the module. The same compliance discipline in three formats.

VIDEO · CODE COMPLIANCE: GETTING KITCHENS APPROVED

0:00

How to design commercial kitchens that pass all four approval gates, from health department to building department.

AUDIO · HOW TO PASS COMMERCIAL KITCHEN INSPECTIONS

0:00 / 0:00

The compliance discipline that prevents failed inspections and costly delays, narrated.

MODULE I · KITCHEN CODE COMPLIANCE GUIDE

Section I Kitchen Code Compliance Guide

Section I Mind Map

MODULE I · FOUNDATIONS

I.1 · WHY COMPLIANCE COMES FIRST

A Kitchen Opens Because It Passes, Not Because It Is Built

Code compliance is not a final-stage correction. It must be designed into the space from the very first layout. A delayed opening costs thousands of dollars a day in lost revenue, idle labor, and extended carrying costs.

The most technically sophisticated kitchen design on paper fails if it cannot survive the approval process. Every piece of equipment, every drain location, every hand sink placement, every exhaust calculation must satisfy the Authority Having Jurisdiction before a permit is issued and an operation begins. Designing to code from the first layout is not optional; it is the foundation that makes every downstream decision defensible.

At **Create The Edge**, code compliance is integrated into the design process, not appended to it. The compliance checklist in this module is the same discipline as the fee defense checklist in Section H and the file discipline checklist in Section K, applied to regulatory approval.

> **THE STRICTER CODE RULE**
>
> When two codes differ, always design to the stricter requirement. The AHJ's interpretation overrides standard text, and local adoptions frequently modify national standards. Always confirm gray areas with the AHJ in writing before the design is committed.

FIGURE I.1.1 · THE FOUR GATES THAT CAN STOP A PROJECT

GATE ONE

Health Department

Sanitation, food flow, and public health. Hand sink placement, clean-to-dirty separation, NSF-listed equipment, finish schedules.

GATE TWO

Building Department

Life safety, structural integrity, and infrastructure. Electrical load, plumbing, fire suppression, mechanical exhaust, ADA clearances.

GATE THREE

Fire Marshal

Life safety systems. Fire suppression alignment, hood certification, egress clearances, suppression nozzle targeting.

GATE FOUR

Accessibility (ADA)

Counter heights, reach ranges, clearances, and accessible routes for all users. Federal requirement on all public and commercial projects.

I.2 · THE FOUR AUTHORITIES

The Authority Having Jurisdiction

The AHJ is the ultimate law of the land. The AHJ's interpretation of any code provision overrides the standard text, which is why confirming gray areas in writing is not optional.

AHJ stands for Authority Having Jurisdiction. It is the local official, agency, or body responsible for enforcing the requirements of a code. In commercial kitchen design, the AHJ is typically the local health department for sanitation, the building department for structural and MEP systems, the fire marshal for life safety, and a separate accessibility inspector for ADA. In some jurisdictions, one agency handles multiple areas.

The critical fact about the AHJ is that its interpretation of any code provision is final. Two kitchens designed identically may face different requirements in different jurisdictions because the AHJ's reading of the applicable code differs. This is why **Create The Edge** confirms gray areas with the AHJ in writing before the design is committed, not after.

FROM THE FILES

A project in one jurisdiction may require hand sinks within 10 feet of every food preparation station; the same code interpreted by a different AHJ may allow 15 feet. Design to what the AHJ in that specific jurisdiction has told you in writing, not to what passed in a different project in a different city. Get the interpretation first, design second.

RULE OF THE HOUSE

Confirm every gray-area code interpretation with the AHJ in writing before finalizing the layout. A verbal approval from a plan reviewer does not bind the AHJ; a written response does. When the answer matters to the design, get it in writing and attach it to the project file.

I.3 · THE CODE STACK

The Commercial Kitchen Code Stack

Codes are layered. National codes establish the baseline. Local adoptions modify them. The AHJ interprets both. When layers conflict, the stricter requirement governs.

FIGURE I.3.1 · THE CODE STACK (TOP TO BOTTOM = MOST TO LEAST AUTHORITY)

AHJ INTERPRETATION

The ultimate authority. Overrides standard text in every conflict. Confirm in writing.

LOCAL ADOPTIONS & AMENDMENTS

Municipal and state modifications to national standards. Municipalities frequently modify adopted editions. Always confirm the current local edition.

STATE CODE ADOPTIONS

State-level adoptions of national codes, often with amendments. The state version governs where no local amendment exists.

NATIONAL CODES (FDA FOOD CODE, IPC, NEC, NFPA)

The national baseline: Food Code for sanitation, IPC for plumbing, NEC for electrical, NFPA for fire safety. Establishes the floor, not the ceiling.

FIGURE I.3.2 · KEY NATIONAL CODES AND THEIR SCOPE

CODE	FULL NAME	WHAT IT GOVERNS
FDA Food Code	U.S. Food & Drug Administration Food Code	Sanitation, food safety, equipment standards, and food-handler practices
IPC	International Plumbing Code	Plumbing systems, drainage, grease interception, water supply
NEC	National Electrical Code (NFPA 70)	Electrical installations, equipment connections, circuit requirements
NFPA 96	Standard for Ventilation Control and Fire Protection	Commercial cooking ventilation, hood requirements, fire suppression
IBC	International Building Code	Structural, occupancy, egress, and ADA baseline requirements
IMC	International Mechanical Code	HVAC, exhaust systems, make-up air, ventilation

I.4 · DUAL-TRACK APPROVAL PATH

The Dual-Track Approval Path

Every commercial kitchen project runs two parallel approval tracks: the Health Department track and the Building Department track. A complete, coordinated plan-review package is the single biggest lever on review speed.

♥

HEALTH DEPARTMENT TRACK

Focus: Sanitation

Evaluates food flow, sanitation, hand sink placement, clean-to-dirty separation, finish schedules (smooth, cleanable surfaces), and NSF-listed equipment. Outcome: Operational Health Permit.

☰

BUILDING DEPARTMENT TRACK

Focus: Life Safety

Evaluates electrical load, plumbing infrastructure, fire suppression, mechanical exhaust, structural connections, and ADA clearances. Outcome: Certificate of Occupancy (CO).

Both tracks run simultaneously from the plan-review stage. A project that clears health but fails building cannot open; neither can a project that clears building but fails health. Coordinating the two packages from the start, ensuring that equipment schedules, utility connections, and spatial layouts satisfy both sets of reviewers, is what produces a first-pass approval rather than a correction cycle that delays opening by weeks.

✓ **BEST PRACTICE**

Submit the health department and building department packages simultaneously where the jurisdiction allows. Staggered submissions mean staggered corrections. A coordinated simultaneous submission compresses the review timeline and reduces the risk of a correction in one track that requires a change in the other.

I.5 · DESIGNING FOR THE HEALTH DEPARTMENT

Designing for the Health Department

The health department evaluates the behavior of the kitchen, looking at sanitation, finish schedules, and cross-contamination risks driven by spatial layout. These are design decisions, not installation decisions.

Health department reviewers read a kitchen layout the same way a food safety inspector reads a kitchen during an operational inspection: they trace the flow of food, staff, and waste to find where cross-contamination risks are highest. A design that fails to establish clear clean-to-dirty separation, or that places hand sinks too far from preparation stations, will receive a correction notice before the first piece of equipment is ordered.

FIGURE I.5.1 · HEALTH DEPARTMENT DESIGN REQUIREMENTS

REQUIREMENT	STANDARD	WHY IT MATTERS
Clean-to-Dirty Separation	Food flow must move from receiving to storage to prep to cooking to service without crossing waste or soiled zones	Cross-contamination of ready-to-eat food is a primary health violation
Hand Sink Placement	Required within a defined radius of every food preparation station; placement is typically 10 to 15 feet, confirmed with the local AHJ	Improper placement is one of the most common correction notices on plan review
Finish Schedules	All food-contact and splash surfaces must be smooth, non-absorbent, and cleanable; coved corners required at floor-wall junctions	Rough or porous surfaces trap bacteria and cannot be sanitized
NSF-Listed Equipment	All food-contact equipment must carry NSF certification (or equivalent)	Proves to the AHJ that the equipment meets baseline commercial sanitation standards
Dedicated Handwashing Sinks	Handwashing sinks must be dedicated to hand washing only; they cannot double as food prep or equipment sinks	Mixing uses creates cross-contamination risk and violates the Food Code

RULE OF THE HOUSE

Create The Edge designs to the most restrictive standard applicable to the project. When the local code requires hand sinks within 10 feet and the FDA Food Code allows 15 feet, design to 10 feet. The stricter requirement is always the correct baseline.

I.6 · NSF CERTIFICATION & SANITATION STANDARDS

NSF Certification and the Baseline of Sanitation

NSF certification is the proof that a piece of commercial equipment meets baseline sanitation standards. Specifying NSF-certified equipment is non-negotiable. It dramatically reduces plan-review rejection risk.

NSF International certifies commercial foodservice equipment to standards that ensure materials, construction, and cleanability meet commercial sanitation requirements. An AHJ reviewing a kitchen layout looks for NSF certification as the baseline indicator that equipment is appropriate for commercial food service. Equipment without certification, or with certification from an unrecognized body, raises immediate questions that slow the approval process.

FIGURE I.6.1 · NSF CONSTRUCTION STANDARDS

STANDARD	REQUIREMENT	WHY IT MATTERS
Coved Corners	Seamless, rounded interior corners at all floor-wall and wall-ceiling junctions in food zones	Prevents debris buildup and allows complete sanitation without gaps
Smooth Welds	Ground and polished joints at all seams on food-contact surfaces	Eliminates bacteria traps in crevices between joined surfaces
Leg Clearances	Minimum 6 inches of clearance beneath equipment for floor cleaning	Allows thorough cleaning under equipment; reduces pest harborage risk
Non-Absorbent Materials	All food-contact surfaces must be smooth, non-porous, and cleanable	Porous materials cannot be sanitized to commercial standards

✓ **BEST PRACTICE**

Specify NSF certification in every equipment description in the MasterSpec. This language signals to the equipment dealer that **Create The Edge** will not accept substitutions that lack the certification, protecting the client from a plan-review correction after the equipment is ordered.

I.7 · PLUMBING PRIORITIES

Plumbing Priorities: Drains, Grease, and Air Gaps

Commercial kitchen plumbing is more complex than residential plumbing because it must manage grease, food waste, and temperature extremes while satisfying both the health department and public works. Getting it right at the layout stage prevents field corrections.

FIGURE I.7.1 · THE PLUMBING TRINITY

1

PRIORITY ONE

Indirect Waste (Air Gap)

Required for all food preparation sinks and ice bins. A direct pipe connection to the sewer is illegal because a downstream clog would push raw sewage directly into the food preparation area. The air gap physically breaks the connection.

2

PRIORITY TWO

Floor Sinks and Drains

Sized and positioned to capture spills and discharge from food equipment. Sloped floors drain toward floor sinks; improper slope creates standing water and health violations.

3

PRIORITY THREE

Grease Interceptor

Mandated by public works to keep fat, oil, and grease out of municipal sewers. Sizing dictates kitchen capacity. Undersized interceptors back up under production load and cause operational failure.

RULE OF THE HOUSE

Every food preparation sink, warewashing sink, and ice bin must discharge through an indirect waste connection, not a direct drain connection. This is the Food Code standard in every jurisdiction **Create The Edge** works in. A direct connection is a code violation that health departments discover immediately on plan review.

⚠ WATCH OUT: GREASE INTERCEPTOR SIZING

Grease interceptors are sized based on the production volume of the kitchen, not the square footage. An interceptor sized for a school servery will fail under the production load of a hospital kitchen of the same area. Size the interceptor for the actual cooking program, not the footprint.

I.8 · EXHAUST HOODS & MAKE-UP AIR

Exhaust Hoods and the Ventilation Matrix

Hood type, coverage, exhaust rate, and make-up air must balance perfectly. An imbalanced ventilation system creates negative pressure, drafts, odor migration, and energy waste, and fails mechanical inspection.

FIGURE I.8.1 · HOOD TYPE SELECTION MATRIX

HOOD TYPE	PRIMARY FUNCTION	FIRE SUPPRESSION	EQUIPMENT TRIGGER
Type I Hood	Removes grease, smoke, and heat from cooking equipment that produces grease-laden vapors	Required. Welded exhaust ducting and listed fire suppression system	Fryers, broilers, griddles, ranges, woks, char-broilers
Type II Hood	Removes heat, moisture, and odors from equipment that does not produce grease vapors	Not required. Standard galvanized ducting	High-temp dishwashers, ovens, pasta cookers (AHJ dependent)

▦ THE EXHAUST-TO-MAKE-UP-AIR BALANCE

Exhaust volume (CFM out) must equal make-up air (CFM in). A kitchen that exhausts more than it supplies creates negative pressure: doors are hard to open, combustion appliances back-draft, and cold air infiltrates from every unsealed gap. **Create The Edge** coordinates the exhaust and make-up air calculations with the MEP engineer during Design Development.

✓ **BEST PRACTICE: VENTLESS EQUIPMENT**

Specify ventless combi ovens where the kitchen layout, budget, or building conditions make hood installation impractical. Ventless models carry UL 710B certification and require no exhaust ducting, eliminating the Type I hood cost ($3,000+ per linear foot) and the make-up air unit. Confirm the ventless certification is current and AHJ-accepted in the specific jurisdiction before committing to the specification.

I.9 · FIRE SUPPRESSION

Fire Suppression: Precision Alignment

Fire suppression nozzles are engineered to target specific cooking surfaces and vat sizes. The suppression system must perfectly match the final cooking lineup. A change to equipment after the system is designed forces a costly permit resubmittal.

Commercial kitchen fire suppression systems are listed systems: each system is certified for specific equipment configurations under a listed arrangement. Nozzle positions, coverage areas, and agent quantities are engineered to the exact equipment lineup. Substituting a 14-inch fryer for an 18-inch fryer invalidates the suppression design and requires a new layout, new calculations, and in most cases a new permit submission.

FIGURE I.9.1 · FIRE SUPPRESSION DESIGN RULES

RULE	WHAT IT MEANS	WHAT VIOLATING IT COSTS
Lock the hotline early	Finalize the cooking equipment schedule before the suppression design begins	Equipment changes after design force a full suppression redesign and permit resubmittal
Nozzle specificity	Each nozzle is positioned and sized for the specific appliance it protects	Improper coverage = failed inspection; potential liability for fire damage

RULE	WHAT IT MEANS	WHAT VIOLATING IT COSTS
No substitutions without re-engineering	Swapping one piece of cooking equipment for another of different size requires new suppression calculations	Delayed occupancy; contractor disputes; potential insurance complications

▦ **RULE OF THE HOUSE**

Lock the hotline equipment schedule before the fire suppression design begins. Moving equipment means moving suppression nozzles and re-engineering the agent calculations. Any cooking equipment change after the suppression design is issued must be communicated to the suppression contractor immediately. **Create The Edge** flags this risk in every proposal scope description.

MODULE I · CRITICAL UTILITIES

I.10 · ELECTRICAL INTEGRITY

Electrical Integrity: The Tag-to-Schedule Match

Every piece of equipment in a commercial kitchen has a nameplate specifying its voltage, phase, amperage, and connection type. The electrical design must match that nameplate exactly. Mismatches create field change orders, delays, and potential equipment warranty voids.

The tag-to-schedule match is the discipline of verifying that the electrical connection specified in the construction documents matches the actual equipment nameplate. A spec that calls for 208V/1-phase and an equipment nameplate that requires 240V/3-phase creates a field change order the day the equipment is delivered. That change order delays the opening, creates contractor disputes, and may void the equipment warranty if the wrong voltage is applied even temporarily.

FIGURE I.10.1 · ELECTRICAL COORDINATION CHECKLIST

VERIFICATION POINT	WHAT TO CONFIRM
Voltage	Nameplate voltage matches the building's available service (208V, 240V, or 480V systems)
Phase	Single-phase vs. three-phase matches both the equipment requirement and the panel capacity
Amperage	Circuit breaker and wire gauge sized for the nameplate amperage, not a rounded estimate
Connection type	NEMA plug configuration or hard-wired connection matches the equipment specification
UL listing	Equipment carries a current UL or ETL listing number confirming factory testing

⚠ WATCH OUT

Equipment manufacturers sometimes change nameplate specifications between the time the equipment is specified and the time it is delivered. Before the electrical rough-in begins, confirm current nameplate data directly from the dealer. A submittal review at the start of construction prevents the field mismatch that causes opening delays.

I.11 · ADA ACCESSIBILITY

ADA Accessibility Requirements

The Americans with Disabilities Act applies to all commercial and public facilities. Counter heights, reach ranges, clearances, and accessible routes are federal requirements, not design preferences.

FIGURE I.11.1 · KEY ADA REQUIREMENTS FOR FOODSERVICE FACILITIES

ELEMENT	ADA STANDARD	DESIGN APPLICATION
Counter Heights	Maximum 34 inches for accessible transaction surfaces	POS stations, serving counters, and customer-side service areas in serveries
Forward Reach	Maximum 48 inches above finished floor; minimum 15 inches	Controls, dispensers, and service items accessible from a wheelchair position
Side Reach	Maximum 54 inches above finished floor	Side-mounted controls and dispensers in serving areas
Knee Clearance	27 inches minimum height, 30 inches minimum width, 19 inches minimum depth	Under-counter clearance at accessible seating and workstations
Accessible Routes	36 inches minimum clear width; 60 inches at turning radius	All paths of travel from entry to seating and service areas

✓ BEST PRACTICE

Confirm ADA requirements with the building department during the Design Development phase. Some jurisdictions adopt state accessibility codes that differ from the federal ADA standard, typically in the stricter direction. Design to the more restrictive of the two, and document the standard used in the drawing set.

I.12 · LAUNDRY & MECHANICAL COMPLIANCE

Laundry and Mechanical Compliance

Commercial laundry facilities carry their own compliance requirements: venting, water supply, drainage, and utility load that differ from foodservice design. When laundry is in scope, it adds a separate compliance layer to the engagement.

FIGURE I.12.1 · COMMERCIAL LAUNDRY COMPLIANCE REQUIREMENTS

SYSTEM	REQUIREMENT	DESIGN IMPACT
Venting	Commercial dryers require dedicated exhaust runs terminated outdoors; no shared duct runs with kitchen exhaust	Separate chase or exterior penetration required; coordinate with mechanical early
Water Supply	Hot and cold water supply sized for the combined flow rate of all washers at simultaneous operation	Undersized supply creates inconsistent wash temperatures and increased cycle times
Drainage	Floor drains positioned beneath washer drain outlets; sized for full discharge during rapid drain cycles	Undersized drains back up; floor slopes must direct water to drains, not doorways
Electrical Load	Commercial washer-extractor and dryer loads are significant; three-phase power commonly required	Panel capacity and electrical service confirmed in SD; coordinate with electrical engineer
ADA Clearances	Front-loading equipment requires knee clearance below; accessible controls at required reach range	Equipment selection affects ADA compliance from the first layout

RULE OF THE HOUSE

When laundry is in scope alongside foodservice, treat it as a separate compliance track from the start. The health department, building department, and MEP coordination requirements for laundry differ from foodservice. Confirm which AHJ has jurisdiction over the laundry space in the specific facility type, as institutional laundry is sometimes governed by a different regulatory body than the kitchen.

MODULE I · INSPECTION READINESS

I.13 · THE PLAN REVIEW PACKAGE

The Health Department Plan Review Package

A complete, coordinated plan-review package is the single biggest lever on review speed. Incomplete *submissions cycle back; complete submissions move forward.*

Health departments review a specific set of documents before issuing a plan approval. Missing any element from the required submission delays the review clock: the reviewer returns the package, the resubmission queue restarts, and the project opening timeline extends. **Create The Edge** assembles the plan review package as a complete submission, not a phased submission, to prevent a correction cycle that costs weeks.

FIGURE I.13.1 · STANDARD PLAN REVIEW SUBMISSION COMPONENTS

DOCUMENT	WHAT IT SHOWS
Equipment Schedule	Full list of all foodservice equipment with model numbers, manufacturer, NSF certification status, and utility connections
Equipment Cut Sheets	Manufacturer specification sheets for each piece of equipment, confirming NSF listing and utility requirements
Kitchen Layout Drawing	Dimensioned floor plan showing equipment placement, hand sink locations, traffic flow, and clean-to-dirty separation
Finish Schedule	All wall, floor, and ceiling finishes in the food prep and storage zones confirming smooth, non-porous, and cleanable materials
Plumbing Diagram	Drain connections, air gaps, grease interceptor, and hand sink locations coordinated with the layout
Ventilation Diagram	Hood coverage, exhaust CFM, make-up air locations, and Type I vs. Type II designations

✓ BEST PRACTICE

Request a pre-submittal conference with the health department plan reviewer before submitting. A 30-minute conversation that surfaces jurisdiction-specific requirements prevents a correction notice after the formal submission clock starts. Most health departments offer this service; **Create The Edge** recommends using it on every project with a new or unfamiliar AHJ.

I.14 · INSPECTION READINESS CHECKLIST

Inspection Readiness Checklist

The inspection readiness checklist confirms that every compliance element is in place before the inspector arrives. A single missing item can result in a failed inspection and a delay to opening that costs the client thousands of dollars per day.

Health Department Readiness

- ☐ All equipment matches the approved equipment schedule exactly (model numbers, NSF certification)
- ☐ Hand sinks installed at approved locations, dedicated to handwashing only
- ☐ Clean-to-dirty separation maintained through equipment layout and traffic flow
- ☐ All food-contact surfaces smooth, non-porous, and cleanable
- ☐ Coved corners installed at all floor-wall junctions in food zones
- ☐ Air gaps installed at all food prep sinks, warewashing, and ice bin drains
- ☐ Grease interceptor installed and accessible for maintenance

Building Department Readiness

- ☐ Type I hoods cover all required cooking equipment; fire suppression system installed and tested
- ☐ Exhaust and make-up air volumes balanced and confirmed by test
- ☐ All electrical connections match approved equipment schedules (voltage, phase, amperage)
- ☐ Equipment has current UL or ETL listing numbers visible on nameplate
- ☐ ADA clearances confirmed at all accessible counters, routes, and controls
- ☐ Floor slopes drain to floor sinks; no standing water in food zones

- [ ] Hotline equipment matches the approved fire suppression layout exactly

> ▦ **RULE OF THE HOUSE**
>
> Run the inspection readiness checklist before requesting the final inspection, not after. A failed inspection restarts the scheduling queue: the inspector returns only when the queue allows, which can add days to weeks to the opening timeline. Inspect yourself first, fix what fails, then call the AHJ.

MODULE I · ASSESSMENT

✓ COMPREHENSIVE TEST

Code, Compliance & Health Department: Assessment

15 questions across the module. Each has one correct answer. A passing score is 12 correct.

1

AHJ stands for:

- **A** Authority Having Jurisdiction
- **B** Approved Health and Janitorial
- **C** Architectural and Housing Journal
- **D** American Hospitality Jurisdiction

2

When two applicable codes conflict on a design requirement, which standard governs?

- **A** The more lenient standard, to minimize client cost
- **B** The stricter standard always governs
- **C** The national standard overrides local codes
- **D** The designer chooses which standard to apply

3

A Type I hood is required above which type of commercial cooking equipment?

- **A** High-temperature dishwashers and standard ovens only
- **B** Pasta cookers and steamers only
- **C** Fryers, broilers, griddles, ranges, woks, and char-broilers that produce grease-laden vapors
- **D** Any equipment that produces heat

4

NSF certification on commercial foodservice equipment primarily proves:

- **A** The equipment was manufactured in the United States
- **B** The equipment meets baseline commercial sanitation standards for materials and cleanability
- **C** The equipment is covered by a commercial warranty
- **D** The equipment qualifies for federal food program use

5

Why must food preparation sinks discharge through an indirect waste connection (air gap) rather than a direct drain connection?

- **A** Direct connections are more expensive to install
- **B** Indirect connections allow faster draining
- **C** A direct pipe connection allows a downstream sewer backup to push raw sewage into the food preparation area
- **D** Local codes require air gaps only on ice machines, not prep sinks

6

The dual-track approval path for commercial kitchens requires simultaneous approval from which two primary authorities?

- **A** The fire marshal and the ADA inspector
- **B** The health department and the building department
- **C** The NSF and the USDA
- **D** The state licensing board and the local zoning authority

7

A grease interceptor is required primarily to:

- **A** Filter cooking odors from the kitchen exhaust
- **B** Reduce the temperature of hot water before it enters the drain
- **C** Capture fat, oil, and grease before it enters the municipal sewer system
- **D** Collect food waste for composting

8

What is the consequence of changing a cooking equipment lineup after the fire suppression system design is complete?

- **A** No consequence if the new equipment is the same type
- **B** The suppression system must be re-engineered and a new permit submission may be required
- **C** The suppression contractor adjusts nozzles without additional paperwork
- **D** Only the hood design needs to change, not the suppression system

9

In the kitchen code stack, the AHJ's interpretation of a code provision:

- **A** Can be overridden by the national code if they conflict
- **B** Only applies to new construction, not renovations
- **C** Overrides the standard text in every conflict and is the final authority
- **D** Is advisory only; the designer may choose to follow the national standard instead

10

The tag-to-schedule match in electrical coordination means:

- **A** Equipment delivery tags must be removed before the health inspection
- **B** The electrical connection specified in the construction documents matches the actual equipment nameplate specifications exactly
- **C** All equipment must be tagged with the contractor's identification number
- **D** Equipment schedule tags must match the floor plan numbering system

11

The maximum counter height for an accessible transaction surface under ADA is:

- **A** 36 inches
- **B** 34 inches
- **C** 42 inches
- **D** 30 inches

12

Exhaust volume (CFM out) in a commercial kitchen ventilation system must equal:

- **A** At least 150% of the make-up air to create positive pressure
- **B** The make-up air volume (CFM in) to maintain balanced pressure
- **C** Half the make-up air volume to allow for natural infiltration
- **D** Whatever the hood manufacturer specifies regardless of make-up air

13

Which of the following is a primary benefit of submitting a complete, coordinated plan-review package to the health department?

- **A** It guarantees automatic approval without a site inspection
- **B** It prevents a correction notice that restarts the review queue and delays the opening timeline
- **C** It allows the project to open before final permits are issued
- **D** It reduces the equipment budget by eliminating NSF requirements

14

When laundry is in scope alongside a foodservice project, Create The Edge treats it as:

- **A** A minor addition that uses the same compliance track as the kitchen
- **B** A separate compliance track with its own venting, water supply, drainage, and utility load requirements
- **C** An optional scope item that does not require AHJ approval
- **D** An architectural element outside Create The Edge's engagement scope

15

The correct time to run the inspection readiness checklist is:

- **A** After the AHJ schedules the final inspection
- **B** Before requesting the final inspection, so that any failures can be corrected before the inspector arrives
- **C** During the construction document phase
- **D** Only if the project has had previous inspection failures

☑ **ANSWER LEGEND · PART 1 OF 2**

Answers · Questions 1 through 8

Do not review until the test is complete.

QUESTION 1

Correct: A · Authority Having Jurisdiction

AHJ = Authority Having Jurisdiction. The local official whose interpretation of any code provision is final and overrides standard text. See I.2.

QUESTION 2

Correct: B · The stricter standard always governs

When two codes conflict, design to the stricter requirement. This applies between national and local codes, and between different national codes governing the same element. See I.3.

QUESTION 3

Correct: C · Fryers, broilers, griddles, ranges, woks, and char-broilers

Type I hoods are required above equipment that produces grease-laden vapors. Type II hoods are for equipment that produces only heat and moisture without grease. See I.8.

QUESTION 4

Correct: B · The equipment meets baseline commercial sanitation standards for materials and cleanability

NSF certification proves to the AHJ that the equipment is appropriate for commercial food service. It covers materials, construction, smooth welds, coved corners, and cleanability. See I.6.

QUESTION 5

Correct: C · A direct pipe connection allows a downstream sewer backup to push raw sewage into the food preparation area

The air gap physically breaks the connection between the drain and the sewer system. A direct connection is illegal under the Food Code in all jurisdictions. See I.7.

QUESTION 6

Correct: B · The health department and the building department

The dual-track path runs health (sanitation, food flow, NSF) and building (life safety, MEP, ADA) simultaneously. Both must approve for the project to open. See I.4.

QUESTION 7

Correct: C · Capture fat, oil, and grease before it enters the municipal sewer system

Grease interceptors prevent FOG from entering municipal sewers, where it creates blockages. Sizing is based on production volume, not kitchen area. See I.7.

QUESTION 8

Correct: B · The suppression system must be re-engineered and a new permit submission may be required

Fire suppression systems are listed to specific equipment configurations. Changing the cooking lineup invalidates the suppression design. Lock the hotline before the suppression design begins. See I.9.

☑ ANSWER LEGEND · PART 2 OF 2

Answers · Questions 9 through 15

The remaining answers with reasoning and sub-module cross-references.

QUESTION 9

Correct: C · Overrides the standard text in every conflict and is the final authority

The AHJ's interpretation is final. No national code, state code, or design standard can override a written AHJ determination. This is why gray-area interpretations must be confirmed in writing. See I.2 and I.3.

QUESTION 10

Correct: B · The electrical connection specified in construction documents matches the actual equipment nameplate exactly

Voltage, phase, amperage, and connection type must match. A mismatch creates a field change order the day equipment is delivered. Confirm current nameplate data from the dealer before rough-in. See I.10.

QUESTION 11

Correct: B · 34 inches

ADA requires accessible transaction surfaces to be no higher than 34 inches above finished floor. This applies to POS stations, serving counters, and customer-facing service areas. See I.11.

QUESTION 12

Correct: B · The make-up air volume (CFM in) to maintain balanced pressure

Exhaust volume must equal make-up air volume. An imbalance creates negative pressure: doors are hard to open, combustion appliances back-draft, and cold air infiltrates through every gap. See I.8.

QUESTION 13

Correct: B · It prevents a correction notice that restarts the review queue and delays the opening timeline

Incomplete submissions cycle back. A complete submission moves forward. The review queue delay from a correction notice can add weeks to the opening timeline. See I.13.

QUESTION 14

Correct: B · A separate compliance track with its own venting, water supply, drainage, and utility load requirements

Laundry compliance differs from foodservice compliance in every utility system. Treating it as a separate track from the start prevents missed requirements at plan review. See I.12.

QUESTION 15

Correct: B · Before requesting the final inspection, so failures can be corrected before the inspector arrives

A failed inspection restarts the scheduling queue. The AHJ returns only when the queue allows, which can add days or weeks to the opening timeline. Inspect yourself first, fix what fails, then call the AHJ. See I.14.

◆ FLASHCARDS · REFERENCE GRID

Twelve Cards · Front and Back

Twelve flashcards covering the key compliance concepts from Section I.

1 AHJ

Authority Having Jurisdiction. The local official whose interpretation of any code provision is final and overrides standard text. Confirm gray areas in writing.

(I.2)

2 The Stricter Code Rule

When two codes conflict, always design to the stricter requirement. The AHJ's interpretation overrides standard text. Local adoptions frequently modify national standards.

(I.3)

3 The Four Approval Gates

Health (sanitation), Building (life safety/MEP), Fire Marshal (suppression), Accessibility (ADA). All four must approve for a project to open.

(I.1)

4 Dual-Track Approval

Health department (sanitation/food flow) and Building department (life safety/infrastructure) run simultaneously. Missing either approval prevents opening.

(I.4)

5 NSF Certification

Proves to the AHJ that equipment meets baseline commercial sanitation standards. Coved corners, smooth welds, 6-inch leg clearance, non-absorbent surfaces.

(I.6)

6 Air Gap (Indirect Waste)

Required at all food prep sinks, warewashing, and ice bins. Physically breaks the connection to the sewer to prevent sewage backflow into food preparation areas.

(I.7)

7 Type I vs. Type II Hood

Type I: grease-laden vapors, fire suppression required. Equipment: fryers, griddles, broilers. Type II: heat and moisture only, no suppression. Equipment: dishwashers, ovens.

(I.8)

8 CFM Balance

Exhaust CFM must equal make-up air CFM. An imbalance creates negative pressure: hard-to-open doors, back-drafting combustion appliances, cold air infiltration.

(I.8)

9 Fire Suppression Lineup Lock

10 Tag-to-Schedule Match

Lock the hotline equipment schedule before the suppression design begins. Any equipment change after the design is issued requires re-engineering and may require a new permit.

(I.9)

Every electrical connection in the construction documents must match the actual equipment nameplate: voltage, phase, amperage, and connection type. Mismatches cause field change orders.

(I.10)

11 ADA Counter Height

Maximum 34 inches for accessible transaction surfaces. Minimum 36-inch clear path width. 60-inch turning radius. Knee clearance: 27H x 30W x 19D inches minimum.

(I.11)

12 Inspect Before Calling the AHJ

Run the inspection readiness checklist before requesting the final inspection. A failed inspection restarts the scheduling queue and can delay opening by days or weeks.

(I.14)

FLASHCARDS · INTERACTIVE

Tap the Card to See the Answer

Twelve cards, one at a time. Read the front, attempt the back from memory, then tap to check.

TERM · TAP TO REVEAL

AHJ

Tap card to flip · Use arrows to navigate

← PREVIOUS 1 of 12 NEXT →

✂ PRINTABLE CUT-OUT STUDY CARDS

Print, Cut, Carry

Twelve cards arranged for printing. Cut along the borders. Keep them on your desk.

AHJ

Authority Having Jurisdiction. Final authority on all code interpretations. Overrides standard text. Confirm gray areas in writing before design is committed.

The Stricter Code Rule

When two codes conflict, design to the stricter requirement. Local adoptions frequently modify national standards; AHJ interpretation is always final.

Four Approval Gates

Health (sanitation), Building (life safety/MEP), Fire Marshal (suppression), ADA (accessibility). All four must approve for a project to open.

Dual-Track Approval

Health department and Building department run simultaneously. A complete, coordinated submission compresses review time and reduces correction cycles.

NSF Certification

Proves equipment meets baseline commercial sanitation. Coved corners, smooth welds, 6-inch leg clearance. Required on all food-contact equipment specifications.

Air Gap (Indirect Waste)

Required at all food prep sinks, warewashing, and ice bins. Physically breaks the sewer connection. A direct connection is a Food Code violation.

Type I vs. Type II Hood

Type I: grease vapors, fire suppression required (fryers, griddles, broilers). Type II: heat/moisture only, no suppression (dishwashers, standard ovens).

CFM Balance

Exhaust CFM = Make-up air CFM. Imbalance creates negative pressure: hard-to-open doors, back-drafting, cold air infiltration, and energy waste.

Fire Suppression Lineup Lock

Lock hotline equipment schedule before suppression design begins. Any change after design requires re-engineering and possibly a new permit submission.

Tag-to-Schedule Match

Electrical connections in construction documents must match equipment nameplates: voltage, phase, amperage, connection type. Mismatches = field change orders.

ADA Counter Height

Maximum 34 inches for accessible transaction surfaces. 36-inch clear path. 60-inch turning radius. Federal requirement on all commercial and public projects.

Inspect Before Calling the AHJ

Run the checklist before requesting final inspection. A failed inspection restarts the scheduling queue and can delay opening by days or weeks.

MODULE I · PRESENTATION

FULL SLIDE DECK

Code and Compliance Blueprint

The complete presentation for this module. Each slide is embedded directly. Scroll through all 15 slides below.